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Purification of Sour Cherry Necrotic Ringspot and Prune Dwarf Viruses[1]

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Sour cherry necrotic ringspot and prune dwarf viruses were purified from cucumber (*Cucumis sativus* L.) cotyledons harvested 3–6 days after inoculation. Infected tissue was homogenized in 0.02 *M* pH 8.0 phosphate buffer containing 0.01 *M* sodium diethyldithiocarbamate and 0.02 *M* sodium thioglycolate. After low speed centrifugation the supernatant was mixed with hydrated calcium phosphate. This adsorbed chloroplastic material and cell debris, which then was removed by low speed centrifugation. Two cycles of low and high speed centrifugation of the clarified homogenate produced clear glassy pellets which were highly infectious. Suspensions of necrotic ringspot virus were composed of round particles about 23 mμ in diameter; prune dwarf virus particles were about 22 mμ in diameter. Purified preparations had ultraviolet absorption spectra typical of nucleoproteins. Healthy cucumber cotyledons processed in the same manner as infected tissue yielded no visible material. Both infectivity and typical particles adsorbed to hydrated calcium phosphate in 0.002 *M* buffer and failed to adsorb in 0.03 *M* buffer. Purified virus in 0.02 *M* phosphate buffer was much more stable than virus in crude extracts. Purified necrotic ringspot virus lost infectivity rather rapidly in distilled water and was quickly inactivated by vigorously shaking the suspension.

INTRODUCTION

Necrotic ringspot and prune dwarf viruses are very commonly found infecting sour cherries (*Prunus cerasus* L.). Their differentiation is important because each may be carried without symptoms in sour cherry. Each has been suspected of being a contributing factor to diseases which may result from the presence of a second virus in infected trees.

Each of these two viruses can be identified by characteristic reactions

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of differential *Prunus* hosts on graft inoculation (Berkeley *et al.*, 1951; Hildebrand *et al.*, 1951), or more rapidly by mechanical inoculation to differential herbaceous hosts (Fulton, 1957a). Neither of these methods of identification, however, eliminates the possibility of confusion with other unrelated viruses which might induce similar symptoms. It appeared that more precise identification would be possible if necrotic ringspot and prune dwarf viruses could be concentrated and purified, their sizes and shapes determined, and the possibility explored of using purified preparations as antigens.

Both viruses are very unstable in undiluted plant extracts and lose most of their infectivity within a few minutes. Various chemicals have been found, however, which are effective in stabilizing infectivity of prune dwarf virus (Fulton, 1957c). A treatment which stabilized both viruses, and which appeared promising as an initial step in purification, consisted of grinding infected tissue in 20–25 volumes of 0.03 *M* pH 8.0 phosphate containing 1–2 volumes of hydrated calcium phosphate (Fulton, 1957b). Most of the chloroplastic material and cell debris adsorbed to the gel-like calcium phosphate and sedimented readily on low speed centrifugation. The viruses remained in the nearly water-clear supernatant and retained their infectivity for several days.

The stability of the viruses in calcium phosphate-treated extracts indicated that some tissue constituent was responsible for the apparent instability of the viruses. It seemed probable that purified virus would be sufficiently stable for use as an antigen in serological investigations. Such material also would be useful in determining the nature of inactivation occurring in tissue extracts.

In previous electron microscopical investigations, Willison *et al.* (1956) presented evidence that necrotic ringspot virus is a sphere about 40 mμ in diameter and that prune dwarf virus is a 30-mμ sphere. Thornberry (1958) reported isolating viruslike particles 80 mμ in diameter from tissue infected with cherry ringspot. These reports appear conflicting, but may not have dealt with the same viruses, since in neither case was the material isolated shown to cause the typical disease in stone fruits.

MATERIALS AND METHODS

The cherry necrotic ringspot and prune dwarf viruses were the same isolates referred to in previous investigations as virus G and virus B, respectively (Fulton, 1957a, 1957c). Virus G caused necrotic ringspot

when inoculated to sour cherry (Fulton, 1958). More recently, virus B has caused typical prune dwarf symptoms when inoculated to Italian Prune (*Prunus domestica* L.) and has caused symptoms nearly identical to illustrations (Willison *et al.*, 1951) of tatter leaf of sweet cherry (*P. avium* L.). Neither virus separately, nor a combination of the two, has caused sour cherry yellows.

Assays showed that extracts of cucumber cotyledons (*Cucumis sativus* L. var. Improved Chicago Pickling) inoculated with necrotic ringspot virus were maximally infectious 3–6 days after inoculation, and that such tissue contained a higher concentration of virus than other hosts investigated. Inoculated cucumber cotyledons were therefore used as a source of necrotic ringspot virus.

Although the time required to reach a maximum concentration in cucumber cotyledons varied seasonally, virus content could be judged roughly from the degree of yellowing of the cotyledons. If these failed to become uniformly yellow within 6 days, or showed only isolated yellow lesions, the virus content was invariably low. In late spring, necrotic ringspot virus usually remained in low concentration in cucumber cotyledons. This appeared to be due to environmental conditions prior to inoculation, as incubation of inoculated plants under controlled conditions failed to result in increased virus concentrations.

The hydrated calcium phosphate (HCP) used to adsorb host material was prepared by adding to a 0.1 *M* solution of Na_2HPO_4 slightly less than an equal volume of 0.1 *M* $CaCl_2$. The flocculent white precipitate settled enough in 4–6 hours to allow one-third or more of the supernatant to be decanted. Distilled water was added and the precipitate resuspended. Decantation and resuspension of the precipitate in distilled water was done fifteen to twenty times to remove soluble salts.

After washing, the HCP was packed by centrifuging at low speed. The final preparation was about the consistency of soft butter and had a water content of 97 %. It was stored in this form at room temperature. The gel-like consistency and the ability to adsorb green plant material were lost after drying and resuspending in water.

Most of the work reported here was done with the necrotic ringspot virus. The method of purification developed for this virus was then applied to the prune dwarf virus in inoculated cucumber cotyledons. The presence of virus in preparations resulting from various steps in the purification procedure was checked by infectivity assays. Local lesion assay hosts used were *Momordica balsamina* L. for necrotic ringspot

virus and *Sesbania exaltata* (Raf.) Cory for prune dwarf virus. Carborundum was used for all inoculations. Virus preparations were diluted with 0.02 or 0.03 *M* pH 8.0 phosphate buffer for inoculation.

RESULTS

Extract Clarification

As reported previously (Fulton, 1957b), adsorption of virus and host material on HCP was differentially affected by the ionic concentration of the homogenate. This was determined by homogenizing infected tissue in 25 volumes of phosphate buffer containing 2.5 volumes of HCP and centrifuging. When the buffer concentration was below about 0.006 *M* both virus and green host material adsorbed to the HCP, leaving a clear, nearly noninfectious supernatant. The presence of virus in the green sediment was demonstrated by resuspending it in 0.04 *M* buffer and centrifuging again. At this buffer concentration, green host material remained adsorbed to HCP and was sedimented. The virus, however, as shown by infectivity tests given in Table 1, appeared in the clear supernatant.

For the purpose of clarification a buffer concentration of 0.03 *M* appeared optimal. After centrifugation of homogenates of infected tissue and HCP in buffer of this concentration a nearly water-clear supernatant was obtained which was usually more infectious than control preparations without HCP. Moreover, necrotic ringspot virus remained infectious in such supernatants for several days at room temperature.

TABLE 1

THE EFFECT OF CONCENTRATION OF PHOSPHATE BUFFER (pH 8.0) ON ADSORPTION AND ELUTION OF NECROTIC RINGSPOT VIRUS FROM HYDRATED CALCIUM PHOSPHATE (HCP)

Buffer concentration of 1:25 virus extract containing 10% (v/v) HCP	Total local lesions on 3 half-leaves inoculated with:	
	Supernatant after sedimenting HCP	HCP eluate[a]
0.015 *M*	558	1
0.01 *M*	451	22
0.006 *M*	3	168
0.003 *M*	1	127

[a] The HCP was resuspended in a volume of 0.04 *M* phosphate buffer twice that of the original extract.

Although stabilization of necrotic ringspot virus was obtained by homogenization with 25 volumes of buffer containing HCP, centrifugal concentration of virus from such large volumes was impractical. When more concentrated homogenates were prepared, containing infected tissue, buffer, and HCP in a w:v:v proportion of 10:15:12, good clarification usually was obtained only if the concentration of the buffer used was below 0.03 *M*. Apparently the amount of soluble electrolytes contributed to the homogenate by the tissue was sufficient to raise the total salt concentration beyond the point where green tissue constituents would adsorb. Lower buffer concentrations resulted in more nearly complete clarification. Good clarification was obtained by homogenizing 10 g of cucumber cotyledons, 15 ml of 0.02 *M* buffer, and 12 ml HCP and centrifuging.

Virus stability often was poor in the more concentrated homogenates, despite what appeared to be excellent removal of green plant constituents. A few preparations retained most of their original infectivity for 6–8 hours; many became nearly completely noninfectious within 2–3 hours. After it was found (Hampton and Fulton, 1959) that necrotic ringspot virus infectivity could be stabilized by including 0.01 *M* sodium diethyldithiocarbamate in extracts, this compound was used to maintain infectivity. Sodium thioglycolate also was included routinely with both viruses, although a mixture of the two compounds was essential in maintaining infectivity of only the prune dwarf virus. Neither sodium diethyldithiocarbamate nor sodium thioglycolate at concentrations up to 0.05 *M* affected adsorption of green material to the HCP.

After virus stabilization was obtained with chemical additives it was found that less HCP was needed for clarification when large tissue fragments were first removed from homogenates by low speed centrifugation. The procedure used for clarification therefore was to homogenize infected tissue in a Waring blendor with a solution of 0.01 *M* sodium diethyldithiocarbamate and 0.02 *M* sodium thioglycolate in 0.02 *M* pH 8.0 phosphate buffer, using 15 ml of solution for each 10 g of tissue. Dow Corning Antifoam A was used to reduce foaming. The homogenate was centrifuged at 2500 rpm for 15 minutes. HCP was added to the supernatant at the rate of 6 ml for each 10 g of original tissue, and the materials were mixed by drawing them into and expelling them from a large syringe fitted with a glass tube instead of a needle. Foaming was reduced by keeping the tip of the tube below the surface of the liquid. This mixture then was centrifuged at 2500 rpm for 15 minutes. One treatment

with HCP removed nearly all green material from the homogenate. Mixing the supernatant a second time with HCP, using half the volume of the first treatment, usually removed an additional small amount of green material, leaving a clear amber supernatant. No attempt was made to keep virus-containing extracts cold, except that homogenization was done in a cold room at 5° C.

Centrifugation

Following clarification, the virus-containing extracts were centrifuged 20 minutes at 10,000 rpm. This apparently only served to remove traces of HCP which were not removed at lower speed. Virus then was pelleted by centrifugation in a Spinco model L ultracentrifuge for 2 hours at 30,000 rpm in a no. 30 rotor. The pellets were composed of a clear glassy portion and a smaller portion of tan-colored insoluble material.

The high speed pellets were resuspended in 0.02 *M* pH 8.0 phosphate buffer by 15–20 minutes' gentle shaking on a mechanical shaker. When resuspended in distilled water, pellets had much lower infectivity than in buffer. The resuspended pellets were centrifuged at 10,000 rpm for 20 minutes; this removed most of the opaque portion of the original pellet. Virus was sedimented again by centrifuging 2 hours in a Spinco no. 40 rotor at 35,000 rpm. The resulting pellets were nearly colorless, and opaque material was reduced to, at most, a small dot in the center of the pellet. Most of this was removed by subsequent centrifugation for 20 minutes at 10,000 rpm.

The final pellet suspensions in 0.02 *M* pH 8.0 phosphate buffer usually represented a ten- to twentyfold concentration of virus, assuming complete recovery. Such suspensions were infectious at dilutions of 1:1000 or more, and comparative assays showed that they contained between 50 and 100 % of the original infectivity. The accuracy of these comparisons was probably low, however, because samples of tissue retained for assays may not have represented accurately the entire lot of tissue. Undiluted purified virus preparations infected stone fruit seedlings readily. Typical symptoms of necrotic ringspot were produced on 4 of 4 sour cherry seedlings inoculated with one preparation of necrotic ringspot virus.

Electron Microscopy

Virus was prepared for electron microscopy by diluting purified preparations 1:10 to 1:30 in water or 0.002 *M* pH 8.0 phosphate buffer.

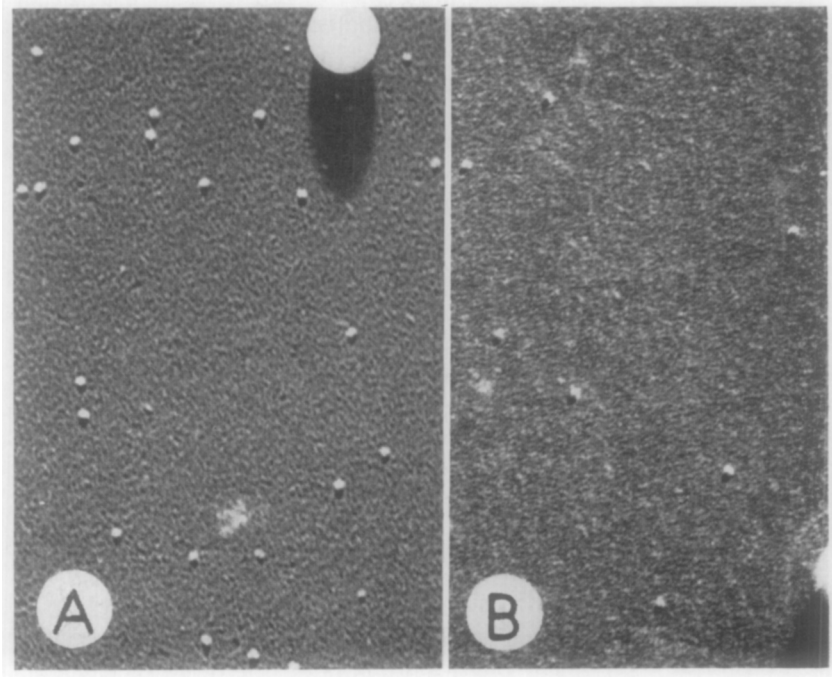


FIG. 1. Electron micrographs of (A) sour cherry necrotic ringspot virus, and (B) prune dwarf virus made from air-dried preparations. Magnification: × 73,000.

Dilutions greater than 1:10 in distilled water were generally unsatisfactory, as particles appeared aggregated in clumps of various sizes. Polystyrene latex spheres of an average diameter of 122.5 mμ were included in all preparations. The diluted virus was sprayed on collodion films supported by electron microscope screen disks and air-dried. After being shadowed with uranium they were viewed in an RCA EMU2 electron microscope.

Necrotic ringspot virus appeared to consist of uniform, apparently round particles (Fig. 1, A). Measurements were made from images of electron micrographs projected on a screen. In comparison with the size of the polystyrene latex spheres the average size of fifty virus particles was 23.5 mμ. There was 25–30% variation in measured widths of individual particles, much of which was probably due to their small size and to indistinctness of the image margins at high magnification. Similar preparations of prune dwarf virus consisted of particles averaging 22.3 mμ in diameter (Fig. 1, B).

The polystyrene spheres in these preparations cast "shadows" 1.5 to 1.7 times their width. The virus particles, on the other hand, cast "shadows" shorter than their widths. Assuming that the virus when suspended in liquid was isodiametric, it appeared to have been much flattened as a result of drying, which would result in apparent diameters greater than the actual diameters.

In an attempt to reduce flattening, virus dilutions were sprayed on electron microscope grids in contact with metal at solid CO_2 temperature and the ice was sublimed by applying a vacuum to the grids in a sealed container. Such frozen-dried necrotic ringspot virus averaged 20.6 mμ in diameter, and prune dwarf virus averaged 20.0 mμ. Even in these preparations, however, the ratio of "shadow" length to particle width was less for virus than for the polystyrene spheres. Whether this discrepancy was due to measurement errors related to the small size of the virus particles on an apparently rough surface or to some other cause is not known.

Relation of Characteristic Particles with Infectivity

Purified preparations appeared to consist almost entirely of a suspension of uniform particles. The reduction in extract volume obtained during purification yielded preparations which were proportionately more infectious than was the same volume of infected tissue extract. It seemed probable therefore that the particles represented the infective units. On a number of electron micrographs, particles larger than the typical size were found. Most of these were rather clearly aggregates of two particles, with their adjacent sides somewhat flattened. Others appeared, although less distinctly, to be clumps of three or more particles.

Other evidence indicated also that the typical particles represented virus. When healthy cucumber cotyledons were processed in the same way as infected tissue, only small opaque pellets were obtained by high speed centrifugation of the HCP-treated supernatant; no clear glassy material was present. No visible material was pelleted by the second high speed centrifugation.

The behavior of infective material in adsorbing to and eluting from HCP in response to changes in ionic strength suggested an additional correlation between the presence of typical particles and the presence of infectivity. Electron microscopic observation of HCP showed that it was composed of very thin platelets which were surprisingly transparent to the electron beam (Fig. 2, A). Purified virus in 0.02 *M* phosphate buffer was diluted 1:10 in distilled water containing HCP. At the ionic strength

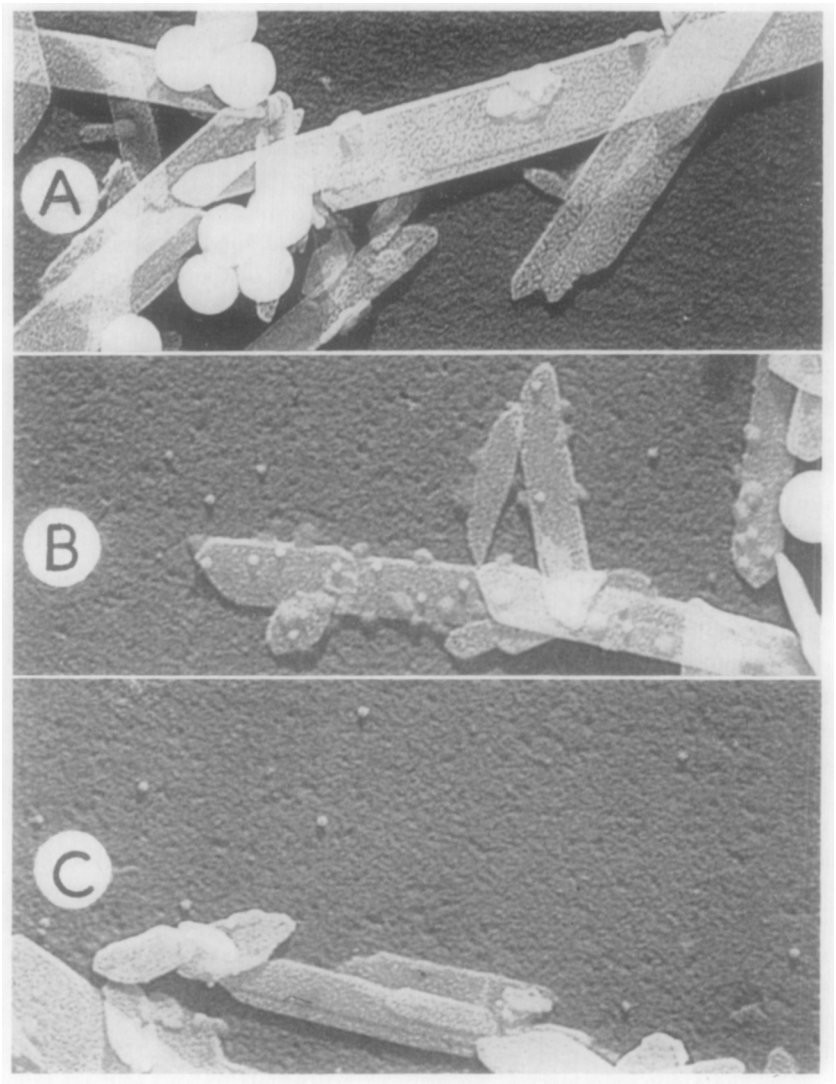


FIG. 2. A. Electron micrograph of microcrystals of hydrated calcium phosphate (HCP) suspended in water and air-dried. B. Particles of necrotic ringspot virus on the surface of and adjacent to HCP. The virus and HCP were suspended in 0.002 *M* buffer, a concentration at which infective material adsorbed to HCP. C. Necrotic ringspot virus and HCP suspended in 0.03 *M* buffer. At this concentration neither infective material nor virus particles adsorbed to HCP. Magnification: × 73,000.

of this mixture (0.002 *M*) infective material adsorbed to the HCP. A similar 1:10 dilution was made in 0.03 *M* phosphate buffer, in which infective material did not adsorb to HCP. Each preparation was sprayed on collodion films, air-dried, shadowed with uranium, and examined in the electron microscope.

The characteristic particles of necrotic ringspot virus behaved in the same way as the infective material, and appeared on or near the HCP in 0.002 *M* buffer (Fig. 2, B). The few particles which did not remain in contact with HCP may have "escaped" when sufficient water evaporated to raise the ionic concentration in the region of the HCP. Particles suspended in 0.03 *M* buffer, on the other hand, remained free of HCP except for the few which might be expected if they were randomly distributed (Fig. 2, C).

Ultraviolet Absorption Spectra of Purified Virus Preparations

Ultraviolet absorption spectra of representative preparations of necrotic ringspot and prune dwarf viruses are shown in Fig. 3. Each virus appeared to give a typical nucleoprotein absorption curve.

Density Gradient Centrifugation

The sedimentation of one preparation of necrotic ringspot virus in a gradient ranging from 5 to 40% sucrose (Brakke, 1958) was compared with a purified preparation of tobacco mosaic virus (TMV). Only one zone was visible in the necrotic ringspot virus preparation after 1, 2, and 3 hours of centrifugation at 25,000 rpm in a Spinco SW 25.1 swinging bucket rotor. The upper boundary of this zone after 1 hour of centrifugation was sharp, but it became somewhat more diffuse after 2 and 3 hours. The rate of sedimentation of the zone was less than half the rate of sedimentation of the TMV zone. The absence of additional zones indicated the absence of major contaminants with other sedimentation characteristics, but it seemed unlikely that the preparation was sufficiently concentrated for a minor contaminant to have produced a visible zone.

Stability of Purified Necrotic Ringspot Virus

Although no systematic study was made of the stability of virus in purified preparations, some observations on necrotic ringspot virus showed that it was more stable than had been inferred from previous studies with crude extracts. Infectivity of crude extracts prepared with 0.1 *M* phosphate buffer was reported (Fulton, 1957a) to be much lower

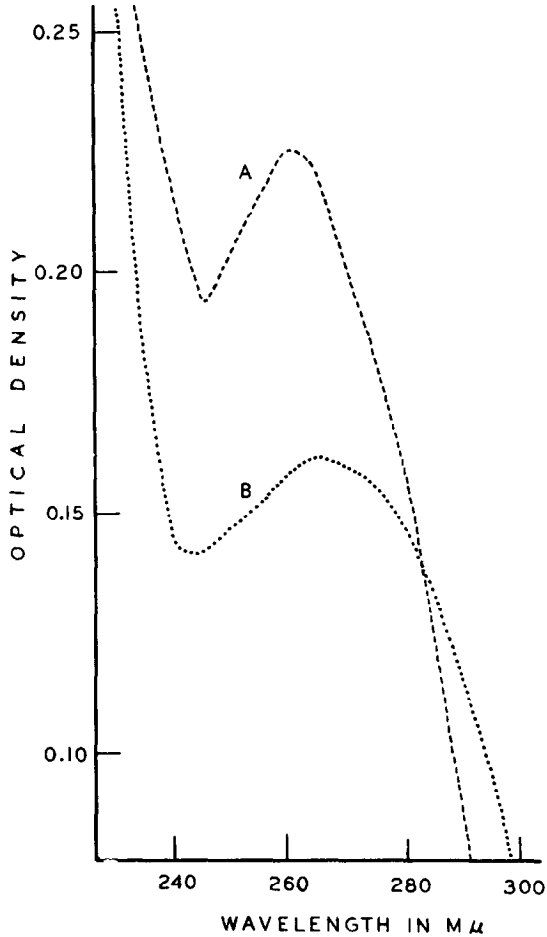


FIG. 3. Ultraviolet absorption spectra of representative preparations of necrotic ringspot virus (*A*), and prune dwarf virus (*B*).

than that of extracts prepared with 0.03 *M* buffer. Purified virus, however, allowed to stand 10 minutes in 0.3 *M* pH 8.0 buffer and then diluted to 0.03 *M* for inoculation was as infectious as a comparable dilution of untreated virus.

Purified virus in 0.02 *M* phosphate buffer at pH 4.4 and pH 5.3 lost no appreciable infectivity in 10–20 minutes. Such preparations were adjusted to pH 8.0 for inoculation.

Purified preparations diluted 1:10 in distilled water to give a buffer concentration of 0.002 *M* and stored for 1 week at 2° were less than 10% as infectious as similar dilutions in 0.03 *M* buffer. Both preparations were further diluted in 0.03 *M* buffer for inoculation. Concentrated purified preparations in 0.02 *M* pH 8.0 phosphate buffer have remained infectious at 2° for 65 days, the longest period tested. It was evident, however, that they were less infectious than when freshly prepared.

Purified virus was very readily inactivated by vigorous shaking to induce foaming. In one trial, virus in 0.03 *M* buffer shaken vigorously by hand for 2 minutes gave 6 lesions on three half-leaves. A preparation identical except for the presence of 1% gelatin induced 54 lesions after shaking. The unshaken control produced 100 lesions. One attempt to resuspend virus pelleted at 80,000 *g* by shaking it vigorously resulted in the appearance of a whitish coagulum and the complete loss of infectivity.

DISCUSSION

In general no data obtained indicated any major differences between the two *Prunus* viruses and the more thoroughly studied viruses of herbaceous plants. The sizes of the *Prunus* viruses place them among the smaller plant viruses. Their instability appears to be the result of a marked sensitivity to certain host components in extracts. Once these are removed or inhibited, stability of the viruses is increased to a point roughly comparable with the stabilities of such viruses as cucumber mosaic or tobacco ringspot.

Stabilization of extract infectivity by sodium diethyldithiocarbamate and sodium thioglycolate raised a question regarding the necessity of using HCP for clarification. A few preparations were treated with quantities of HCP insufficient to remove all green color. Two cycles of high and low speed centrifugation failed to remove all the green color from the pellets. It seemed clear that the relatively simple and rapid clarification with HCP facilitated subsequent centrifugal concentration and purification.

The possibility of using HCP for other steps in addition to initial clarification was investigated. Its use in a chromatographic column was impractical because of extremely slow flow rates. Attempts to concentrate virus by adsorbing it on HCP from dilute extracts and then eluting with small quantities of buffer gave poor yields. There appeared to be a rather rapid loss of virus infectivity at ionic concentrations at which virus adsorbed to HCP. Furthermore, quantitative elution of adsorbed

virus required successive washings with a total volume of buffer approaching the volume of the original extract.

Although the yield of virus from a known weight of tissue was not determined, it was quite evident from the small size of the pellets that the amount of virus obtained was less than that obtained with many other plant viruses. Infectivity comparisons did not indicate any major loss of virus during purification. It appeared therefore that the two *Prunus* viruses exert their rather severe pathological effects in cucumber without reaching a concentration comparable with that of many plant viruses.

The diameters of necrotic ringspot and prune dwarf viruses found in this investigation were markedly less than those reported by Willison *et al.* (1956) or by Thornberry (1958). The differences appear too large to be due to variations in technique, in spite of numerous possible sources of error in making precise electron microscopic measurements. It is possible that different viruses were measured in each case. Thornberry's preparations were reportedly noninfective and could have contained particles which were a product of disease rather than its cause. Willison *et al.* determined sizes on the basis of an increase in the proportion of particles in certain size ranges in infected tissue as compared to healthy tissue. These also could have been products of deranged host metabolism rather than virus. Or they could have been aggregates of smaller particles.

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